The Coordinate Pythagorean Theorem from Planar Isotropy:

An Axiomatic Derivation with Lie-Theoretic Extensions

AI-generated manuscript: **ChatGPT (GPT-6, High thinking effort)**
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Abstract

We derive the coordinate Pythagorean identity from a primitive length function, axial calibration, mirror symmetry, and a transitive rigid one-parameter family of linear planar motions containing an oriented quarter-turn. The matrix of the quarter-turn and the normal form of the family are derived from algebraic hypotheses. Rigidity then implies a determinant-one identity, and rotating a vector onto the horizontal axis yields $a^2 + b^2 = c^2$, without taking the Cartesian diagonal-distance formula as a premise. The main proof is group-theoretic and does not require differentiability. The original Lie-theoretic treatment, using the generator J and classification of invariant quadratic forms, is retained as a supplementary derivation. The manuscript is AI-generated; the motivating request to seek an original proof is not itself evidence of historical novelty.

1 Introduction

Origin, AI authorship, and research prompt. This manuscript was written by OpenAI's ChatGPT, using the **GPT-6 model with High thinking effort**, following a prompt from **Alessandro Odasso** asking whether ChatGPT could find an *original proof of the Pythagorean theorem*. Odasso supplied the initiating research question; the prose and proposed mathematical derivation in the original draft were generated by AI. The word *original* describes the objective of the prompt, *not* an independently established claim of novelty, priority, or publication-level originality. The mathematical argument should be evaluated on its explicit assumptions and deductions, independently of the system that generated the text. The present revision also incorporates later Lean-formalized developments: the group-theoretic proof is now presented first, and the differentiable Lie-theoretic viewpoint is retained as an extension. Thus this version is not merely the unchanged August draft with a new title.

Many familiar proofs of the Pythagorean theorem use geometric dissection, similarity, or trigonometric/algebraic identities. When extending the theorem to analytic geometry, standard presentations frequently define the Cartesian metric distance via $d(P_1, P_2) = \sqrt{\Delta x^2 + \Delta y^2}$, introducing potential circularity if used to evaluate diagonal segment lengths.

We distinguish two levels. The principal theorem uses a linear one-parameter action with a geometrically specified quarter-turn, primitive length congruence, mirror symmetry, and axis reachability. A subsequent Lie-theoretic specialization studies the infinitesimal generator and its invariant quadratic forms. Both routes are conditional results and neither defines the primitive length by a Euclidean square root.

2 Primary axiomatic and group-theoretic derivation

Let $L : \mathbb{R}^2 \to \mathbb{R}$ be a primitive geometric length function, not defined by $\sqrt{x^2+y^2}$. We state the exact hypotheses used for the principal result:

1. **Axial calibration:** $L(z,0) = |z|$ for every real z.
2. **Mirror symmetry:** $L(x,-y) = L(x,y)$ for all x, y.
3. **Primitive oriented quarter-turn:** a linear map Q obeys $Q(1,0) = (0,1)$ and $Q^2 = -I$.
4. **Linear group action:** a family of linear maps F_t satisfies $F_{s+t} = F_s \circ F_t$ for all $s, t \in \mathbb{R}$, and some F_θ equals Q.
5. **Rigidity:** $L(F_t(x,y)) = L(x,y)$ for every t, x, y.
6. **Axis reachability:** for the chosen vector (a,b), there exist $T \in \mathbb{R}$ and $z \geq 0$ with $F_T(a,b) = (z,0)$.

The last two conditions are geometric assumptions, not conclusions of linearity alone. The main proof does not assume differentiability.

Lemma 2.1 (The metric-free quarter-turn normal form)**.** *The quarter-turn satisfies $Q(x,y) = (-y,x)$. Every F_t has the form*

$$F_t(x,y) = \big(u(t)x - v(t)y,\ v(t)x + u(t)y\big) \tag{1}$$

for suitable real functions u, v.

Proof. The square relation gives $Q(0,1) = Q^2(1,0) = (-1,0)$. Linearity yields $Q(x,y) = (-y,x)$. Since $F_tQ = F_{t+\theta} = F_{\theta+t} = QF_t$, the arbitrary matrix $F_t = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ commutes with $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Thus $q = -r$ and $s = p$. Taking $u(t) = p$, $v(t) = r$ proves (1). □

Lemma 2.2 (Rigidity eliminates radial scaling)**.** *For every t, the coefficients in* (1) *satisfy $u(t)^2 + v(t)^2 = 1$.*

Proof. Fix t and abbreviate $u = u(t)$, $v = v(t)$. Rigidity and axial calibration imply $L(u,v) = L(F_t(1,0)) = L(1,0) = 1$. Reflection symmetry gives $L(u,-v) = 1$. But (1) gives $F_t(u,-v) = (u^2+v^2, 0)$. Hence

$$|u^2+v^2| = L(u^2+v^2,0) = L(u,-v) = 1.$$

As the sum of two squares is nonnegative, it equals 1. No quadratic distance formula was assumed for L. □

Theorem 2.1 (Group-theoretic coordinate Pythagorean theorem)**.** *Under the six stated hypotheses, let $c = L(a,b)$. Then $a^2 + b^2 = c^2$.*

Proof. Choose T, z with $z \geq 0$ and $F_T(a,b) = (z,0)$. Let $u = u(T)$, $v = v(T)$. The normal form yields

$$ua - vb = z, \qquad va + ub = 0,$$

and Lemma 2.2 gives $u^2 + v^2 = 1$. By a direct algebraic expansion,

$$(ua - vb)^2 + (va + ub)^2 = (u^2+v^2)(a^2+b^2) = a^2 + b^2.$$

The left-hand side is z^2. Independently, rigidity and axial calibration identify

$$c = L(a,b) = L(F_T(a,b)) = L(z,0) = z.$$

Consequently $a^2 + b^2 = z^2 = c^2$. □

Remark 2.1 (What is and is not assumed)**.** *The coordinate form of a quarter-turn, the normal form of F_t, and the determinant-one identity are derived, not postulated. However, linearity, the half-turn property, presence of the quarter-turn in an abelian one-parameter action, mirror symmetry, rigidity, and reachability are substantive hypotheses. Lean verifies this implication, not the independence or minimality of these axioms.*

3 Lie-theoretic specialization: the original approach

We formalize the relationship between primitive Euclidean geometry and Cartesian coordinates through two axioms.

Axiom 3.1 (Orthogonal Coordinate Calibration)**.** *The two coordinate directions correspond to perpendicular geometric directions. Furthermore, any two points in $\mathbb{R}^2$ whose coordinate difference vector is $(L, 0)$ or $(0, L)$ have a primitive Euclidean geometric separation equal to $|L|$.*

Axiom 3.2 (Rigid Rotational Congruence and Transitivity)**.** *Continuous rotation about the origin $C(0, 0)$ is a length-preserving rigid motion belonging to the one-parameter flow group*

$$G = \{R(t) = e^{tJ} : t \in \mathbb{R}\}, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{2}$$

Furthermore, for any two rays emanating from C, there exists a transformation $R(T) \in G$ mapping one ray onto the other.

3.1 Classification of invariant quadratic forms under G

We establish that the infinitesimal generator J forces the form of the Euclidean quadratic invariant up to an overall scalar multiplier.

Lemma 3.1 (Classification of Nonzero Symmetric Quadratic Invariants)**.** *Let $Q(\mathbf{r}) = \mathbf{r}^T M \mathbf{r}$ be a nonzero symmetric quadratic form generated by $M = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$. Then Q is invariant under the rotational flow G if and only if $M = \lambda I$ for some scalar $\lambda \neq 0$.*

Proof. By definition, Q is invariant under $R(t) = e^{tJ}$ if and only if $R(t)^T M R(t) = M$ for all $t \in \mathbb{R}$. Differentiating at $t = 0$ yields the infinitesimal condition

$$J^T M + M J = 0. \tag{3}$$

Uniqueness. Since $J^T = -J$, equation (3) reduces to $MJ = JM$. Direct multiplication gives

$$MJ = \begin{pmatrix} \beta & -\alpha \\ \gamma & -\beta \end{pmatrix}, \qquad JM = \begin{pmatrix} -\beta & -\gamma \\ \alpha & \beta \end{pmatrix}. \tag{4}$$

Equating corresponding entries forces $\beta = 0$ and $\alpha = \gamma$. Setting $\alpha = \gamma = \lambda$ gives $M = \lambda I$. Since $Q \neq 0$, we have $\lambda \neq 0$.

Existence. Conversely, suppose $M = \lambda I$ with $\lambda \neq 0$. Then

$$J^T M + M J = \lambda(J^T + J) = \lambda(-J + J) = 0. \tag{5}$$

Differentiating $Q(\mathbf{r}(t))$ along any integral curve $\dot{\mathbf{r}} = J\mathbf{r}$ yields

$$\frac{d}{dt} Q(\mathbf{r}(t)) = \mathbf{r}^T (J^T M + M J) \mathbf{r} = 0. \tag{6}$$

Thus every quadratic form $Q(x, y) = \lambda(x^2 + y^2)$ with $\lambda \neq 0$ is conserved along the flow G. □

Remark 3.1. *The identities $R(t)^T R(t) = I$ and $\det R(t) = 1$ follow from $J^T = -J$ and $\operatorname{tr} J = 0$. Thus G is a one-parameter subgroup of $SO(2)$ generated by J.*

3.2 The original dynamical and congruence proof

Theorem 3.1 (Coordinate Pythagorean Theorem)**.** *In a calibrated orthogonal coordinate frame with primitive length congruence, the length-preserving flow e^{tJ}, and ray transitivity, let $\triangle CAB$ be a right triangle with leg lengths $|CA| = a$ and $|AB| = b$, and hypotenuse length $|CB| = c$. Then*

$$a^2 + b^2 = c^2.$$

Proof. Choose the origin at C, the positive x-axis along CA, and the positive y-axis parallel to AB.

By the coordinate-calibration axiom, the coordinate axes represent orthogonal geometric directions. Because $|CA| = a$, vertex A has coordinates $(a, 0)$. Because AB is parallel to the y-axis with $|AB| = b$, vertex B has coordinates (a, b). The hypotenuse CB has primitive geometric length c.

By the rigid-congruence and transitivity axiom, there exists $R(T) = e^{TJ} \in G$ that maps the ray CB onto the positive x-axis, placing $B(a, b)$ at $B'(r, 0)$ with $r > 0$. We evaluate the terminal axial coordinate r by two channels.

Channel 1: Dynamical conservation. By the invariant-quadratic-form lemma, $Q(x, y) = \lambda(x^2 + y^2)$ ($\lambda \neq 0$) is conserved along the trajectory from (a, b) to $(r, 0)$:

$$Q(a, b) = Q(r, 0) \quad \Longrightarrow \quad \lambda(a^2 + b^2) = \lambda(r^2 + 0^2). \tag{7}$$

Dividing by $\lambda \neq 0$ gives

$$a^2 + b^2 = r^2. \tag{8}$$

Channel 2: Geometric congruence. By the rigid-congruence and transitivity axiom, rigid rotation preserves primitive geometric length, so $|CB'| = |CB| = c$. By the coordinate-calibration axiom, the segment from $(0, 0)$ to $(r, 0)$ has geometric length r. Therefore

$$r = c. \tag{9}$$

Combining equations (8) and (9) yields

$$a^2 + b^2 = c^2. \tag{10}$$

□

Remark 3.2 (Foundational Scope)**.** *This derivation proves the Pythagorean theorem relative to an underlying Euclidean congruence structure and a specified rotational flow, rather than deriving metric geometry from purely algebraic primitives. Its contribution lies in showing that once continuous rotational flow and primitive length congruence are logically decoupled from coordinate metrics, rotational symmetry forces the $x^2 + y^2$ quadratic form up to scale, while congruence identifies its terminal axial coordinate with primitive geometric length without assuming the Cartesian distance formula.*

4 Conclusion

The coordinate Pythagorean identity follows algebraically from a primitive length function, rotational group structure and rigid congruence without assuming an a priori Cartesian diagonal-distance formula. The infinitesimal Lie-theoretic method remains valuable for understanding the generator and invariant quadratic forms, but is not required for the principal proof. Both formulations remain conditional on their explicitly stated geometric hypotheses.

Companion formalization (added in the disclosure revision). The Lean 4/Mathlib development verifies the group-theoretic proof in `pythagoras_from_metric_free_quarter_turn`, the derived quarter-turn and flow shape, and the separate infinitesimal-generator results. It is publicly available at `https://github.com/alexodas/pythagorean-isotropy-lean`. Compilation verifies the conditional theorems as formalized; it does not establish historical novelty or the minimality of the geometric assumptions.

References

[1] B. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed., Graduate Texts in Mathematics, Springer, 2015.

[2] N. Jackson and C. Johnson, "Five or Ten New Proofs of the Pythagorean Theorem," *The American Mathematical Monthly*, vol. 131, no. 9, pp. 739–752, 2024.